Homework #4

Math 104: Intro to Analysis

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Contents

0 Various Theorems

0.1 Monotone Decrease I

$a_n = \frac{1}{n(\log n)(\log\log n)^p}$ for $n \in \mathbb{N}$ is monotonically decreasing if $p > 0$.

Proof. Let $p > 0$. We know $n > 0$ since $n \in \mathbb{N}$. This also means $n \geq 1$. Logarithms are monotone, as mentioned in Rudin (p.63). Therefore, $\log n$ and by extension $\log\log n$ and $(\log\log n)^p$ are monotonically increasing. Therefore, $n(\log n)(\log\log n)^p$ is monotonically increasing. On the other hand, this also means that $\frac{1}{n(\log n)(\log\log n)^p}$ is monotonically decreasing. OEΔ

0.2 Monotone Decrease II

$b_n = \frac{1}{n(\log 2)(\log(n)+\log(\log 2))^p}$ for $n \in \mathbb{N}$ is monotonically decreasing if $p > 0$.

Proof. This uses a lot of what was used in Monotone Decrease I. n is monotone increasing. $\log(n)$ is monotone increasing and $\log(\log(2))$ is constant, so $\log(n) + \log(\log(2))$ and by extension $(\log(n) + \log(\log(2)))^p$ if $p > 0$ are monotone increasing. $\log(2)$ is constant, so $n(\log 2)(\log(n) + \log(\log 2))^p$ is monotone increasing if $p > 0$. Therefore, $\frac{1}{n(\log 2)(\log(n)+\log(\log 2))^p}$ is monotone decreasing if $p > 0$. OEΔ

0.3 Inverese Triangle Inequality

For $a, b, c \in (X, d)$, $|d(p, q) - d(q, r)| \leq d(p, r)$.

Proof. From the triangle inequality, we have two equations.

$$d(p, q) \leq d(p, r) + d(q, r) \tag{1}$$
$$d(q, r) \leq d(p, r) + d(p, q) \tag{2}$$

We can subtract $d(q, r)$ and $d(p, q)$ from them respectively.

$$d(p, q) - d(q, r) \leq d(p, r) \tag{3}$$
$$d(q, r) - d(p, q) \leq d(p, r) \tag{4}$$

Reverse the latter.

$$d(p, q) - d(q, r) \geq -d(p, r) \tag{5}$$

By the definition of the absolute value, we can say that $|d(p, q) - d(q, r)| \leq d(p, r)$. OEΔ

0.4 Finite Times Natural is Finite

A finite number times any natural number is finite.

Proof by Induction. Let $x \in \mathbb{R}$ be finite, so $x \neq \pm\infty$.

Base case: $n = 1$. If $n = 1$, then $n \cdot x = 1 \cdot x = x$. Since x is finite, this means the base case holds.

Inductive step: suppose that nx is finite. Take $(n + 1)x$. The distributive property says that $(n + 1)x = nx + x$. Since both x and nx are finite, $nx + x$ is finite. Ergo, $(n + 1)x$ is finite. OEΔ

0.5 Decreasing Value is Greater than Alternating Partial Sums

If $a_k \geq 0$ for all $k \in \mathbb{N}$ and a_k is decreasing, then $\left|\sum_{k=N}^{m}(-1)^{k-1}a_k\right| \leq a_N$ for all $m, N \in \mathbb{N}$ where $m > N$.

Proof. Let $a_k \geq 0$ and $a_k \geq a_{k+1}$ for every $k \in \mathbb{N}$. Let $N \in \mathbb{N}$, and define $T_r = \sum_{j=0}^{r}(-1)^j a_{N+j}$. We will prove by induction that $0 \leq T_{2q+1} \leq T_{2q} \leq a_N$ for $q \geq 0$.

Start with the base case. $T_0 = a_N$, so $0 \leq T_0 \leq a_N$. Since a_k is decreasing, $a_N - a_{N+1} \geq 0$. Since $a_k \geq 0$, $a_N \geq a_N - a_{N+1} \geq 0$, so $a_N = T_0 \geq T_1 \geq 0$. Therefore, the base case is satisfied.

Now for the inductive step. Suppose that for some $q \geq 0$, $0 \leq T_{2q+1} \leq T_{2q} \leq a_N$. Start with T_{2q+2}. Therefore, $T_{2q+2} = T_{2q} - (a_{N+2q+1} - a_{N+2q+2})$. Since the sequence is decreasing, $a_{N+2q+1} \geq a_{N+2q+2}$, so $a_{N+2q+1} - a_{N+2q+2} \geq 0$, so $T_{2q} \geq T_{2q+2}$. Also, since $T_{2q+1} \geq 0$ and $a_{N+2q+2} \geq 0$, $T_{2q+1} + a_{N+2q+2} = T_{2q+2} \geq 0$.

Next, T_{2q+3}. Since $T_{2q+3} = T_{2q+2} - a_{N+2q+3}$ and $a_{N+2q+3} \geq 0$, $T_{2q+3} \leq T_{2q+2}$. Furthermore, $T_{2q+3} = T_{2q+1} + a_{N+2q+2} - a_{N+2q+3}$. Since $a_{N+2q+2} \geq a_{N+2q+3}$, $a_{N+2q+2} - a_{N+2q+3} \geq 0$. Therefore, because $T_{2q+1} \geq 0$, $T_{2q+1} + a_{N+2q+2} - a_{N+2q+3} = T_{2q+3} \geq 0$.

Consolidating, this means that $a_N \geq T_{2q+3} \geq T_{2q+4} \geq 0$. This completes the induction and means that for all $m, N \in \mathbb{N}$ where $m \geq N$, $0 \leq T_m \leq a_N$. By the definition of T_m, for all $m, N \in \mathbb{N}$ where $m \geq N$, $0 \leq \sum_{k=N}^{m}(-1)^{k-1}a_k \leq a_N$. Ergo, since $a_N \geq 0$, $\left|\sum_{k=N}^{m}(-1)^{k-1}a_k\right| \leq a_N$. OEΔ

0.6 Union of Bounded Sets in Bounded

The union of finitely many bounded sets is bounded.

Proof. Let $\{S_n\}_{n=1}^{t}$ be a finite collection of bounded sets. Call the upper and lower bounds of set S_n, U_n and L_n respectively and call their sets $\{U_n\}$ and $\{L_n\}$. Call $\max\{U_n\} = U$ and $\min\{L_n\} = L$. Let $s \in \bigcup_{n=1}^{t} S_n$. There exists a set $S_k \in \{S_n\}$, such that $s \in S_k$. Therefore, $L_k \leq s \leq U_k$. By the definition of the maximum and minimum of the set, $L \leq L_k$ and $U_k \leq U$. Transistively, this means that for any $s \in \bigcup_{n=1}^{t} S_n$, $L \leq s \leq U$. Ergo, the union $\bigcup_{n=1}^{t} S_n$ is bounded. OEΔ

1 Problem 1

For which values of $p \in \mathbb{R}$ are the following series convergent or divergent? Prove your answers rigorously, using only tests we explicitly covered in class. *Hint:* for both of these series you should use something with the name "Cauchy" in it.

(a) $\displaystyle\sum_{n=3}^{\infty} \frac{1}{n(\log n)(\log\log n)^p}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$.

1.1 Solution (a)

It converges if $p > 1$ and diverges otherwise.

Proof. Since $2 < e$, $\log(2) < \log(e) = 1$. Furthermore, through the use of a calculator, $\log 2 = 0.6931$ and $\log\log 2 = -0.3665$.

Let $p \leq 0$. This means $a_n = \frac{(\log\log n)^q}{n(\log n)}$ for some $q \geq 0$ such that $q = -p$. $\log n$ is monotonically increasing, so by extension so is $\log\log n$. Therefore, since $\log n$ is monotonically increasing and unbounded, so is $\log\log n$ and so is $(\log\log n)^q$. Therefore, for some $N_0 \in \mathbb{N}$, for all $n \geq N_0$, $\log\log n \geq 1$ and $(\log\log n)^q \geq 1$. Therefore if $n \geq N_0$, $\frac{(\log\log n)^q}{n\log n} \geq \frac{1}{n\log n}$. Recall that if $a_n \geq d_n \geq 0$ for $n \geq N_0$ and if $\sum d_n$ diverges, then $\sum a_n$ diverges (Rudin, p.60). In Rudin (p.62), it is proven that $\sum_{n=2}^{\infty} \frac{1}{n\log n}$ diverges. Therefore, $\sum_{n=3}^{\infty} \frac{(\log\log n)^q}{n\log n}$ diverges.

Going forward, we will therefore only operate on cases where $p > 0$. If $p > 0$, there exists $N_0 \in \mathbb{N}$ such that for all $n \geq N_0$, all terms that make up $\frac{1}{n(\log n)(\log\log n)^p}$ are greater than 0, so $\frac{1}{n(\log n)(\log\log n)^p} > 0$. Furthermore, it is established earlier in this document that $\frac{1}{n(\log n)(\log\log n)^p}$ is monotonically decreasing.

We will use the Cauchy Condensation Test, which says that for monotonically decreasing, positive a_n, $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} 2^n a_{2^n}$ converges. Consider it in this case.

$$a_n = \frac{1}{n(\log n)(\log\log n)^p} \tag{6}$$

$$b_n = 2^n a_{2^n} = \frac{2^n}{2^n(\log 2^n)(\log\log 2^n)^p} = \frac{1}{n(\log 2)(\log(n\log 2))^p} = \frac{1}{n(\log 2)(\log(n) + \log(\log 2))^p} \tag{7}$$

Now, we can use how $\frac{1}{n\log 2(\log(n)+\log(\log 2))^p}$ is positive and monotonically decreasing to use the Cauchy Condensation Test again.

$$b_n = \frac{1}{n(\log 2)(\log(n) + \log(\log 2))^p} \tag{8}$$

$$c_n = 2^n b_{2^n} = \frac{2^n}{2^n(\log 2)(\log(2^n) + \log(\log 2))^p} = \frac{1}{(\log 2)(n\log(2) + \log(\log 2))^p} \tag{9}$$

Here, we have two cases: either $p > 1$ or $p \leq 1$.

First consider the case where $p \leq 1$ where $n \in \mathbb{N}$ and $n \geq 3$. Now, we know that $\log(2) < 1$, so $n\log(2) < n$. Therefore, since $\log\log 2 < 0$, $n\log(2) + \log\log(2) < n$ and by extension $(n\log(2) + \log\log(2))^p < n^p$. Therefore, $\frac{1}{(n\log(2)+\log\log(2))^p} > \frac{1}{n^p}$. If we sum this for $n \geq 3$, we find that $\sum_{n=3}^{\infty} \frac{1}{(n\log(2)+\log\log(2))^p} > \sum_{n=3}^{\infty} \frac{1}{n^p}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges to ∞ for $p \leq 1$, we can equally say that $\sum_{n=3}^{\infty} \frac{1}{n}$ diverges to ∞ under the same conditions. Recall again that if $a_n \geq d_n \geq 0$ for $n \geq N_0$ and if $\sum d_n$ diverges, then $\sum a_n$ diverges (Rudin, p.60). Since $\sum_{n=3}^{\infty} \frac{1}{n^p}$ diverges to ∞ for $p \leq 1$, $\sum_{n=3}^{\infty} \frac{1}{(n\log(2)+\log\log(2))^p}$ also diverges to ∞ for $p \leq 1$. Therefore, by the Cauchy Condensation Test, if $0 < p \leq 1$, $\sum_{n=3}^{\infty} \frac{1}{n(\log n)(\log\log n)^p}$ diverges.

Now we consider the case where $p > 1$ for $n \in \mathbb{N}$ and $n \geq 3$. I will contend right now that for all $n \in \mathbb{N}$ where $n \geq 3$, $n\log(2) + \log\log(2) > \frac{n}{2}$ and will prove it inductively. The base case is that $n = 3$. With the help of our calculator, we can find that $3\log(2) + \log\log(2) = 1.7129 > \frac{3}{2}$, so our base case holds. For the inductive step, we assume that $n\log(2) + \log\log(2) > \frac{n}{2}$. Our calculator tells us that $\log(2) = 0.693$, so $\log(2) > \frac{1}{2}$. Since $a > b$ and $c > d$ implies that $a + c > b + d$, $n\log(2) + \log\log(2) + \log(2) > \frac{n}{2} + \frac{1}{2}$. Therefore, $(n+1)\log(2) + \log\log(2) > \frac{n+1}{2}$, so the inductive step holds. If $p > 1$, then $(n\log(2) + \log\log(2))^p > \left(\frac{n}{2}\right)^p$. Therefore, $\frac{1}{(n\log(2)+\log\log(2))^p} < \frac{2^p}{n^p}$.

We know that if $p > 1$, then $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent and by extension $\sum_{n=3}^{\infty} \frac{1}{n^p}$. Since a series is still a sequence of partial sums, 2^p is finite if $p \in \mathbb{R}$ and $p > 1$, and a convergent sequence times a finite constant converges, $2^p \sum_{n=3}^{\infty} \frac{1}{n^p} = \sum_{n=3}^{\infty} \frac{2^p}{n^p}$ still converges. If $|a_n| \leq c_n$ for all $n \in \mathbb{N}$ and $n \geq N_0$ for some $N_0 \in \mathbb{Z}$ and $\sum c_n$ converges, then a_n converges (Rudin, p.60). Therefore, since $\frac{1}{(n\log(2)+\log\log(2))^p}$ is made up of only terms that are always positive, it is always positive, so $\left|\frac{1}{(n\log(2)+\log\log(2))^p}\right| = \frac{1}{(n\log(2)+\log\log(2))^p}$. Therefore, $\left|\frac{1}{(n\log(2)+\log\log(2))^p}\right| \leq \frac{2^p}{n^p}$ for all $n \in \mathbb{N}$ and $n \geq 3$. Since $\sum_{n=3}^{\infty} \frac{2^p}{n^p}$ converges, $\sum_{n=3}^{\infty} \frac{1}{(n\log(2)+\log\log(2))^p}$ also converges. Therefore, by the Cauchy Condensation Test, if $p > 1$, $\sum_{n=3}^{\infty} \frac{1}{n(\log n)(\log\log n)^p}$ converges.

Ergo, $\sum_{n=3}^{\infty} \frac{1}{n(\log n)(\log\log n)^p}$ converges if $p > 1$ and diverges otherwise. OEΔ

1.2 Solution (b)

Start with the case that $p > 1$.

Proof. If $p > 1$, then we take the absolute value of each term. We find that $\left|\frac{(-1)^{n-1}}{n^p}\right| = \frac{1}{n^p}$. In Problem 2, we prove that if $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges. Since $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$, then so does $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$. OEΔ

Now the case that $p < 0$.

Proof. If $p < 0$, then for $q = -p$ (so $q > 0$), $\frac{1}{n^p} = n^q$. Since $n \in \mathbb{N}$ and $q > 0$, $n^q > 1$, so the sequence does not converge to 0. Therefore, since a series converges only if its sequence converges to zero and $\frac{1}{n^p}$ does not converge to 0, $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges if $p < 0$. OEΔ

Now the case that $p = 0$.

Proof. If $p = 0$, then for all p, $\frac{1}{n^p} = 1$. This turns the sequence $\frac{(-1)^{n-1}}{n^p}$ into $(-1)^{n-1}$, which does not converge to 0 and whose limit does not exist. Therefore, since a series converges only if its sequence converges to zero and $\frac{1}{n^p}$ does not converge to 0, $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges if $p = 0$. OEΔ

Last, we consider the case where $0 < p \leq 1$.

Proof. According the Rudin (p.57), $\frac{1}{n^p}$ converges to 0 if $p > 0$. Since $\frac{1}{n} > 0$ for all $n \in \mathbb{N}$, $\frac{1}{n^p} > 0$. We proved earlier that if $a_k \geq 0$ for all $k \in \mathbb{N}$ and a_k is decreasing, then $\left|\sum_{k=N}^{m} (-1)^{k-1} a_k\right| \leq a_N$ for all $m, N \in \mathbb{N}$ where $m > N$. Therefore, since $\frac{1}{n^p} > 0$ for all $n \in \mathbb{N}$ and $\frac{1}{n^p}$ is decreasing, $\left|\sum_{k=N}^{m} \frac{(-1)^{k-1}}{k^p}\right| \leq a_n$ for all $m, n \in \mathbb{N}$ where $m > n$. Since $\frac{1}{n^p}$ converges to 0 if $p > 0$, for all $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n| < \varepsilon$. Since $\left|\sum_{k=n}^{m} \frac{(-1)^{k-1}}{n^p}\right| \leq a_n$ for any $n \in \mathbb{N}$, $\left|\sum_{k=n}^{m} \frac{(-1)^{k-1}}{n^p}\right| \leq |a_n|$. By extension, from the convergence of $\frac{1}{n^p}$, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $m \geq n \geq N$, $\left|\sum_{k=n}^{m} \frac{(-1)^{k-1}}{n^p}\right| < \varepsilon$. By the Cauchy Criterion, this means that the series converges. OEΔ

2 Problem 2

A series $\sum_{n=1}^{\infty} a_n$ is said to be *absolutely convergent* if $\sum_{n=1}^{\infty} |a_n|$ is convergent. Prove that every absolutely convergent series is convergent. *Hint:* use the Cauchy criterion.

Proof. Suppose that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. This means that $\sum_{n=1}^{\infty} |a_n|$ is convergent and by extension Cauchy. By the Cauchy Criterion, this means that for all $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that if $m \geq n \geq N$, $|\sum_{k=n}^{m} |a_k|| \leq \varepsilon$.

Recall the triangle inequality for the usual metric: if $a, b, c \in \mathbb{R}$, $|a - c| \leq |a - b| + |b - c|$. Since $|a - c| = |a - b + b - c|$, $|a - b + b - c| \leq |a - b| + |b - c|$. Suppose that for some $a_k = a - b$ and $a_{k+1} = b - c$. Therefore, $|a_k + a_{k+1}| \leq |a_k| + |a_{k+1}|$. Since $m - n$ is finite, we can extend this process to every $k \in [n, m-1] \cap \mathbb{N}$ to conclude that $|\sum_{k=n}^{m} a_k| \leq \sum_{k=n}^{m} |a_k|$. By the absolute value, $|\sum_{k=n}^{m} a_k| \geq 0$, meaning $\sum_{k=n}^{m} |a_k| \geq 0$ and therefore $\sum_{k=n}^{m} |a_k| = |\sum_{k=n}^{m} |a_k||$. This means that $|\sum_{k=n}^{m} a_k| \leq |\sum_{k=n}^{m} |a_k||$.

Transistively, we can say from our earlier convergence statement that for all $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that if $m \geq n \geq N$, $|\sum_{k=n}^{m} a_k| \leq \varepsilon$. Therefore, by the Cauchy Criterion, $\sum_{n=1}^{\infty} a_n$ is convergent. Ergo, if a series is absolutely convergent, it is convergent. OEΔ

3 Problem 3

Let X, Y be sets and $f : X \to Y$ a function.

(a) Show that, for $E, F \subset Y$ we have $f^{-1}(E \cup F) = f^{-1}(E) \cup f^{-1}(F)$ and $f^{-1}(E \cap F) = f^{-1}(E) \cap f^{-1}(F)$.

(b) Show that, for $E, F \subset X$, we have $f(E \cup F) = f(E) \cup f(F)$. Is it also always true that $f(E \cap F) = f(E) \cap f(F)$? Prove or give a counterexample.

(c) Let $E \subset X$. Show that $E \subset f^{-1}(f(E))$. Show that equality holds for all E if and only if f is injective.

(d) Let $E \subset Y$. Show that $f(f^{-1}(E)) \subset E$. Show that equality holds for all E if and only if f is surjective.

3.1 Solution (a)

First we prove that $f^{-1}(E \cup F) = f^{-1}(E) \cup f^{-1}(F)$.

Proof. Begin with $f^{-1}(E \cup F)$. If $x \in f^{-1}(E \cup F)$, then $f(x) \in E \cup F$. Therefore, $f(x) \in E$ or $f(x) \in F$. If $f(x) \in E$, then $x \in f^{-1}(E)$. If $f(x) \in F$, then $x \in f^{-1}(F)$. Therefore, since either $f(x) \in E$ or $f(x) \in F$, $x \in f^{-1}(E)$ or $x \in f^{-1}(F)$. Therefore, $x \in f^{-1}(E) \cup f^{-1}(F)$. This means $f^{-1}(E \cup F) \subset f^{-1}(E) \cup f^{-1}(F)$.

For the reverse direction, begin with $x \in f^{-1}(E) \cup f^{-1}(F)$. This means that either $x \in f^{-1}(E)$ or $x \in f^{-1}(F)$. Without loss of generality, suppose that $x \in f^{-1}(E)$. This means that, $f(x) \in E$. Quite logically, this means that $f(x) \in E$ or $f(x) \in F$. Therefore, $f(x) \in E \cup F$. Therefore, $x \in f^{-1}(E \cup F)$. Concluding, $f^{-1}(E) \cup f^{-1}(F) \subset f^{-1}(E \cup F)$.

Since $f^{-1}(E \cup F) \subset f^{-1}(E) \cup f^{-1}(F)$ and $f^{-1}(E) \cup f^{-1}(F) \subset f^{-1}(E \cup F)$, $f^{-1}(E \cup F) = f^{-1}(E) \cup f^{-1}(F)$. OEΔ

Next, we prove that $f^{-1}(E \cap F) = f^{-1}(E) \cap f^{-1}(F)$.

Proof. Begin with $f^{-1}(E \cap F)$. If $x \in f^{-1}(E \cap F)$, then $f(x) \in E \cap F$. Therefore, $f(x) \in E$ and $f(x) \in F$. Since $f(x) \in E$, $x \in f^{-1}(E)$. Since $f(x) \in F$, $x \in f^{-1}(F)$. Therefore, $x \in f^{-1}(E) \cap f^{-1}(E)$. Therefore, $f^{-1}(E \cap F) \subset f^{-1}(E) \cap f^{-1}(F)$.

Suppose $x \in f^{-1}(E) \cap f^{-1}(F)$. Then, $x \in f^{-1}(E)$ and $x \in f^{-1}(F)$. Since $x \in f^{-1}(E)$, $f(x) \in E$. Since $x \in f^{-1}(F)$, $f(x) \in F$. Therefore, $f(x) \in E \cap F$. Therefore, $x \in f^{-1}(E \cap F)$. Therefore, $f^{-1}(E) \cap f^{-1}(F) \subset f^{-1}(E \cap F)$.

Since $f^{-1}(E \cap F) \subset f^{-1}(E) \cap f^{-1}(F)$ and $f^{-1}(E) \cap f^{-1}(F) \subset f^{-1}(E \cap F)$, $f^{-1}(E \cap F) = f^{-1}(E) \cap f^{-1}(F)$. OEΔ

3.2 Solution (b)

First, we prove that $f(E \cup F) = f(E) \cup f(F)$.

Proof. Suppose that $x \in X$ and $f(x) \in f(E \cup F)$. This means that $x \in E \cup F$. Therefore, either $x \in E$ or $x \in F$. Without loss of generality, suppose that $x \in E$. Therefore, $f(x) \in f(E)$. Using the same steps, we can show that $f(x) \in f(F)$. Therefore, $f(x) \in f(E) \cup f(F)$. This means that $f(E \cup F) \subset f(E) \cup f(F)$.

Suppose that $x \in X$ and $f(x) \in f(E) \cup f(F)$. This means that either $f(x) \in f(E)$ or $f(x) \in f(F)$. Therefore either $x \in E$ or $x \in F$. This means that $x \in E \cup F$. Therefore, $f(x) \in f(E \cup F)$. This means that $f(E) \cup f(F) \subset f(E \cup F)$.

Since and $f(E \cup F) \subset f(E) \cup f(F)$ and $f(E) \cup f(F) \subset f(E \cup F)$, $f(E \cup F) = f(E) \cup f(F)$. OE$\Delta$

Now we disprove that $f(E \cap F) = f(E) \cap f(F)$.

Proof by Counterexample. Take the case of $E = (0, \infty)$ and $F = (-\infty, 0)$ where $f : \mathbb{R} \to \mathbb{R}$ and $f(x) = |x|$. Taking the intersection, $E \cap F = \varnothing$. Therefore, $f(E \cap F) = \varnothing$.

Now, applying the function to E and F, we find $f(E) = (0, \infty)$ and $f(F) = (0, \infty)$. Taking the intersection of the two, we find $f(E) \cap f(F) = (0, \infty)$.

Since $(0, \infty) \neq \varnothing$, $f(E \cap F) \neq f(E) \cap f(F)$. OE$\Delta$

3.3 Solution (c)

First, we prove that $E \subset f^{-1}(f(E))$.

Proof. Let $x \in E$. Therefore, $f(x) \in f(E)$. Since $f(x) \in f(E)$ and $f^{-1}(f(E)) = \{x \in X \Big| f(x) \in f(E)\}$, $x \in f^{-1}(f(E))$. Therefore, $E \subset f^{-1}(f(E))$. OEΔ

Next we prove that if f is injective, then $E = f^{-1}(f(E))$.

Proof. Injective means that if $a, b \in E$, $f(a) = f(b)$ implies that $a = b$. Suppose that f is injective. Recall the definitions that $f(E) = \Big\{f(x) \Big| x \in E\Big\}$ and $f^{-1}(f(E)) = \Big\{x \Big| f(x) \in f(E)\Big\}$.

Take any element in $x \in f^{-1}(f(E))$. By the definition of $f^{-1}(f(E))$, this means that $f(x) \in f(E)$. By the definition of $f(E)$, there exists $y \in E$ such that $f(x) = f(y)$. Since f is injective, this means that $x = y$. Therefore, since $y \in E$ and $x = y$, $x \in E$. This means that if $x \in f^{-1}(f(E))$, $x \in E$. Therefore, $f^{-1}(f(E)) \subset E$.

We proved earlier that $E \subset f^{-1}(f(E))$. Therefore, $f^{-1}(f(E)) \subset E$ and $E \subset f^{-1}(f(E))$. Combining them, that means $E = f^{-1}(f(E))$. OEΔ

Third, we prove that if $E = f^{-1}(f(E))$, then f is injective.

Proof. Recall the definitions that $f(E) = \Big\{f(x) \Big| x \in E\Big\}$ and $f^{-1}(f(E)) = \Big\{x \Big| f(x) \in f(E)\Big\}$.

Suppose that $E = f^{-1}(f(E))$ and $x, y \in X$. Let $x \in E$ and $y \notin E$, so $x \neq y$. Therefore, if $x \in f^{-1}(f(E))$ and $y \notin f^{-1}(f(E))$. This also means that $f(x) \in f(E)$. Additionally, $f(y) \notin f(E)$, because if $f(y) \in f(E)$, then by definition $y \in f^{-1}(f(E))$, which is a contradiction. Therefore, $x \neq y$ implies $f(x) \neq f(y)$. By contrapositive, this means that $f(x) = f(y)$ implies $x = y$. Ergo, f is injective. OEΔ

Last, we consolidate.

Proof. We know that if f is injective, then $E = f^{-1}(f(E))$. We also know that if $E = f^{-1}(f(E))$, then f is injective. Therefore, f is injective if and only if $E = f^{-1}(f(E))$. OEΔ

3.4 Solution (d)

First we prove that $f(f^{-1}(E)) \subset E$.

Proof. Take $y \in f(f^{-1}(E))$. Therefore, there exists some element $x \in f^{-1}(E)$ such that $f(x) = y$. Since $x \in f^{-1}(E)$, then $f(x) \in E$. Since $y = f(x)$, $y \in E$ for any $y \in f(f^{-1}(E))$. Ergo, $f(f^{-1}(E)) \subset E$. OEΔ

Second, we prove that if f is surjective, then $f(f^{-1}(E)) = E$.

Proof. Take any $x \in E$. Since f is surjective, for every element $z \in E$, there is an element $y \in X$ such that $f(y) = z$. Therefore, there is and element $y \in X$ such that $f(y) = x$. By the definition of $f^{-1}(E)$, this means that $y \in f^{-1}(E)$. Furthermore, by the definition of $f(f^{-1}(E))$, since $y \in f^{-1}(E)$, $f(y) \in f(f^{-1}(E))$. Since $f(y) = x$, $x \in f(f^{-1}(E))$. Therefore, $x \in E$ implies $x \in f(f^{-1}(E))$. Ergo, $E \subset f(f^{-1}(E))$.

We have established that $E \subset f(f^{-1}(E))$. We have established that $f(f^{-1}(E)) \subset E$. Ergo, $f(f^{-1}(E)) = E$. OEΔ

Third, we prove that if $f(f^{-1}(E)) = E$, then f is surjective.

Proof. Suppose that for all E, $f(f^{-1}(E)) = E$. This means that if $y \in E$ then $y \in f(f^{-1}(E))$. Since $y \in f(f^{-1}(E))$, there exists $x \in f^{-1}(E)$ where $f(x) = y$. Since $f^{-1}(E) \subset X$, $x \in X$. Therefore, for any $y \in E$, there exists $x \in X$ such that $f(x) = y$. Ergo, f is surjective within E for all $E \subset Y$.

Suppose that $E = Y$. That would mean that f is surjective in E and as such surjective in Y. Therefore, f is surjective over Y. OEΔ

Last, we consolidate.

Proof. We know that if f is surjective, then $E = f(f^{-1}(E))$. We also know that if $E = f(f^{-1}(E))$, then f is surjective. Therefore, f is surjective if and only if $E = f(f^{-1}(E))$. OEΔ

4 Problem 4

Let (X, d_X) and (Y, d_Y) be metric spaces.

(a) A function $f : X \to Y$ (not necessarily continuous) is said to be *locally bounded* if, for each $p \in X$, there is an $r > 0$ so that $f(B_r(p))$ is a bounded set in Y. Show that every locally bounded function on a *compact* metric space is bounded (i.e. $f(X)$ is a bounded set in Y).

(b) A function $f : X \to Y$ is said to be *locally constant* if, for each $p \in X$, there is an $r > 0$ so that f is constant on $B_r(p)$. Show that every locally constant function on a *connected* metric space is constant. *Hint:* show that $f^{-1}(\{y\})$ is clopen in X for any $y \in Y$

4.1 Solution (a)

Proof. Let $f : X \to Y$ be locally bounded and let (X, d_X) be compact. Since (X, d_X) is compact, every open cover of X has a finite subcover. By the definition of a locally bounded function, for each $p \in X$, there is an $r_p > 0$ so that $f(B_{r_p}(p))$ is bounded in Y. Create a collection out of every $B_{r_p}(p)$ and call it P. Since every open ball contains its center and open balls are open, the collection is an open cover of X. Since X is compact, there is a cover of X made up of a finite number of elements in P, which we can call F.

Create a new set called G defined as $G = \{f(S)\}_{S \in F}$. Since every element in F is an open ball, applying f to every open ball returns a bounded set in Y, and F is finite, G is a finite collection of bounded sets in Y. Since the union of a finite number of bounded sets is bounded, the union of all elements in G, $H = \bigcup_{S \in G} S$, is a bounded set. Furthermore, since H is a collection of bounded sets in Y, $H \subset Y$.

Now, we prove that if $f(x) \in f(X)$, then $f(x) \in H$. Take any element $f(x) \in f(X)$. By the definition of $f(X)$, $x \in X$. Since F is a cover of X, there exists $T \in F$ such that $x \in T$. Therefore, $f(x) \in f(T)$. Since $H = \bigcup_{S \in G} S = \bigcup_{S \in F} f(S)$, $f(T) \subset H$. $f(x) \in f(T)$ and $f(T) \subset H$ means $f(x) \in H$. Therefore, $f(x) \in f(X)$ implies $f(x) \in H$. Therefore, $f(X) \subset H$.

Lastly, a few facts to put everything together. Since H is bounded and $f(X) \subset H$, $f(X)$ is bounded. Since $H \subset Y$ and $f(X) \subset H$, $f(X) \subset Y$. Therefore, $f(X)$ is a bounded set in Y. Ergo, every locally bounded function on a compact metric space is bounded. OEΔ

4.2 Solution (b)

Proof by Contradiction. Let $f : X \to Y$ be locally constant and let (X, d_X) and (Y, d_Y) be metric spaces. Furthermore, let (X, d_X) be connected, meaning the only clopen sets in (X, d_X) are $\varnothing$ and X.

Suppose for contradiction that f is not constant. This means that there exists a set $V \subset Y$ with more than one element where $V = f(X)$. For all $v \in V$, let X_v be the set of elements $p \in X$ where $f(p) = v$. Since f is locally constant, for each $p \in X$, there is an $r > 0$ such that for all $q \in B_r(p)$, $f(p) = f(q)$. If $p \in X_v$, then $q \in X_v$ and by extension $B_r(p) \subset X_v$. Therefore, for all $v \in V$, X_v is open.

Since f is a well-defined function, for every $p \in X$, there exists $f(p) \in Y$. Therefore, for every $p \in X$, there exists $v \in V$ where $f(p) = v$. Therefore, for every $p \in X$, there exists $v \in V$ such that $p \in X_v$. Therefore, $X = \bigcup_{v \in V} X_v$ and for any $w \in V$, $X_w^c = X \backslash X_w = \bigcup_{v \in V \backslash \{w\}} X_v$.

Take any $w \in V$. Since $V = f(X)$, $w \in f(X)$, meaning there is some element $x \in X$ where $f(x) = w$. Therefore, $X_w \neq \varnothing$. Furthermore, f is not constant, so $X_w \neq X$. We already proved that for every $v \in V$, X_v is open. Therefore, since the union of open sets is open, $\bigcup_{v \in V \backslash \{w\}} X_v$

is also open. This means that X_w^c is open. Since a set is closed if and only if its complement is open, X_w is closed. Therefore, X_w is both open and closed, or clopen. This contradicts the fact that X is connected, since the only clopen subsets of connected set X are $\varnothing$ and X. Ergo, $f(X)$ is constant. OEΔ

5 Problem 5

Using the $\epsilon - \delta$ definition of continuity directly, prove that the following functions are continuous.

(a) $f : \mathbb{R} \to \mathbb{R}; \quad f(x) = 104x.$

(b) $f : \mathbb{R} \to \mathbb{R}; \quad f(x) = x^2.$

5.1 Solution (a)

Proof. For a function to be continuous, it must be continuous at every point in its domain. Recall the $\epsilon - \delta$ definition of continuous at a point: for all $\varepsilon > 0$, there exists some $\delta > 0$ such that $d_X(x,p) < \delta$ implies $d_Y(f(x), f(p)) < \varepsilon$. In the real numbers and the usual metric, this means that $|x-p| < \delta$ implies that $|f(x) - f(p)| < \varepsilon$.

Let $\varepsilon > 0$ and $\delta = \frac{\varepsilon}{104} > 0$. Suppose that $|x-p| < \delta$. This means that $|x-p| < \frac{\varepsilon}{104}$. Multiplying everything by 104, we find that $\varepsilon > 104\,|x-p| = |104x - 104p|$. Since $f(x) = 104x$, $|f(x) - f(p)| < \varepsilon$. Putting things together, this means that for any $\varepsilon > 0$, there exists some $\delta > 0$ such that $|x-p| < \delta$ implies $|f(x) - f(p)| < \varepsilon$. Ergo, $f(x) = 104x$ is continuous. OEΔ

5.2 Solution (b)

Proof. Let $\varepsilon > 0$. Take any $x, p \in \mathbb{R}$. By the definition of the absolute value, $|x+p| \geq 0$. There are two conditions from here, that either $x = p$ or $x \neq p$.

Suppose $x = p$. This means that $|x-p| = 0$, so for any $\delta > 0$, $\delta > |x-p|$. Furthermore, since $|f(x) - f(p)| = \left|x^2 - p^2\right| = |x-p|\,|x+p|$, $\left|x^2 - p^2\right| = |x-p|\,|x+p| = 0\,|x+p| = 0$. Therefore, $\varepsilon > |f(x) - f(p)|$, so if $x = p$, the $\epsilon - \delta$ definition is satisfied.

Suppose $x \neq p$, so $|x+p| \neq 0$, so $|x+p| > 0$ by the definition of the absolute value. This means $|x+p| > 0$. Let $\delta = \frac{\varepsilon}{|x+p|}$ and suppose that $|x-p| < \delta$. Therefore, $|x-p| < \frac{\varepsilon}{|x+p|}$ Multiply both sides by $|x+p|$ to find that $|x-p|\,|x+p| < \varepsilon$. Therefore, $|(x-p)(x+p)| < \varepsilon$. Therefore, $\left|x^2 - p^2\right| < \varepsilon$. This means $|f(x) - f(p)| < \varepsilon$. Therefore, the $\epsilon - \delta$ definition is satisfied.

The $\epsilon - \delta$ definition is satisfied in all cases. Ergo, $f(x) = x^2$ is continuous. OEΔ

6 Problem 6

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases}$. Using the $\epsilon - \delta$ definition of continuity,

(a) prove that f is not continuous at any point $x \neq 0$;

(b) prove that f *is* continuous at $x = 0$.

6.1 Solution (a)

This is an altered version of Thomae's Function. We consider this in two cases: $x \in \mathbb{Q}\backslash\{0\}$ and $x \neq \mathbb{Q}$.

First we prove that f is not continuous at x if $x \in \mathbb{Q}\backslash\{0\}$.

Proof. Take $\varepsilon = \frac{|f(x)|}{2}$. Take any $\delta > 0$ such that $|x - p| < \delta$. Since the irrational numbers are dense in the real numbers, between any two real numbers, there is an irrational number. This means that there is an irrational number p where $x - \delta < p < x + \delta$. This means that $-\delta < p - x < \delta$, and by extension $|p - x| = |x - p| < \delta$.

Now take $|f(x) - f(p)|$. Since $x \in \mathbb{Q}$ and $p \notin \mathbb{Q}$. Therefore, $|f(x) - f(p)| = |x - 0| = |x|$. Therefore, since $\varepsilon = \frac{|f(x)|}{2} = \frac{|x|}{2}$, $|x| = |f(x) - f(p)| \geq \varepsilon$. Ergo, by the $\epsilon - \delta$ definition, f is not continuous at $x \in \mathbb{Q}\backslash\{0\}$. OEΔ

Next, we prove that f is not continuous at x if $x \in \mathbb{Q}$.

Proof. Take any $\delta > 0$ such that $|x - p| < \delta$. Since the rational numbers are dense in the real numbers, there is a non-zero rational number p where $x - \delta < p < x + \delta$. Therefore, $|x - p| < \delta$.

Now, look at the epsilon-side. Since p is rational but x is irrational, $|f(x) - f(p)| = |p - 0| = |p|$. If $\varepsilon > |p| > 0$, then $\varepsilon > |f(x) - f(p)|$. However, if $0 < \varepsilon \leq |p|$, then $\varepsilon \leq |f(x) - f(p)|$. Ergo, by the $\epsilon - \delta$ definition of continuous, f is not continuous at x if $x \in \mathbb{Q}$. OEΔ

If necessary, we consolidate this.

Proof. We know that f is not continuous at x if $x \in \mathbb{Q}$. We know f is not continuous at x if $x \in \mathbb{Q}\backslash\{0\}$. Therefore, f is not continuous at x if $x \in \mathbb{R}\backslash\{0\}$. Ergo, f is not continuous at any point $x \neq 0$. OEΔ

6.2 Solution (b)

Proof. Let $\varepsilon > 0$. Suppose that $\delta = \varepsilon$. Assume that $|x - p| < \delta$. This means that $|p - x| < \delta$. Since $x = 0$, $|p| < \delta$. Since $\delta = \varepsilon$, this also means $|p| < \varepsilon$.

Now, we apply this to the function. Suppose that $p \notin \mathbb{Q}$. This means that $|x - p| = |0 - 0| = |0| = 0$. By definition, we know that this works, since $\varepsilon > 0$. Suppose that $p \in \mathbb{Q}$. This means that $|x - p| = |p - 0| = |p|$. Earlier in the proof, we established that $|p| < \varepsilon$, so it holds.

Ergo, f is continuous at $x = 0$. OEΔ